

Student Performance Analysis System

1. **Aim :** To design and implement a Student Performance Analysis System in Java using Object-Oriented Programming concepts to calculate total marks, average, percentage, grade, highest and lowest marks, pass/fail status, and overall academic performance.
2. **Description :** The Student Performance Analysis System is a Java application designed for colleges to analyze a student's academic performance.

The system accepts the student's roll number, name, and marks in five subjects. It uses an array to store marks and performs various calculations such as total marks, average marks, highest marks, lowest marks, and percentage.

The system assigns a grade according to the percentage and determines whether the student has passed or failed. String operations such as trim(), toUpperCase(), and length() are used to format and process the student's name.

The Math.round() method is used to round the percentage to two decimal places. The program also displays a performance remark based on the student's grade.

3. Software Used :

Software:

- Java JDK
- Eclipse IDE / Command Prompt
- Windows Operating System

Hardware:

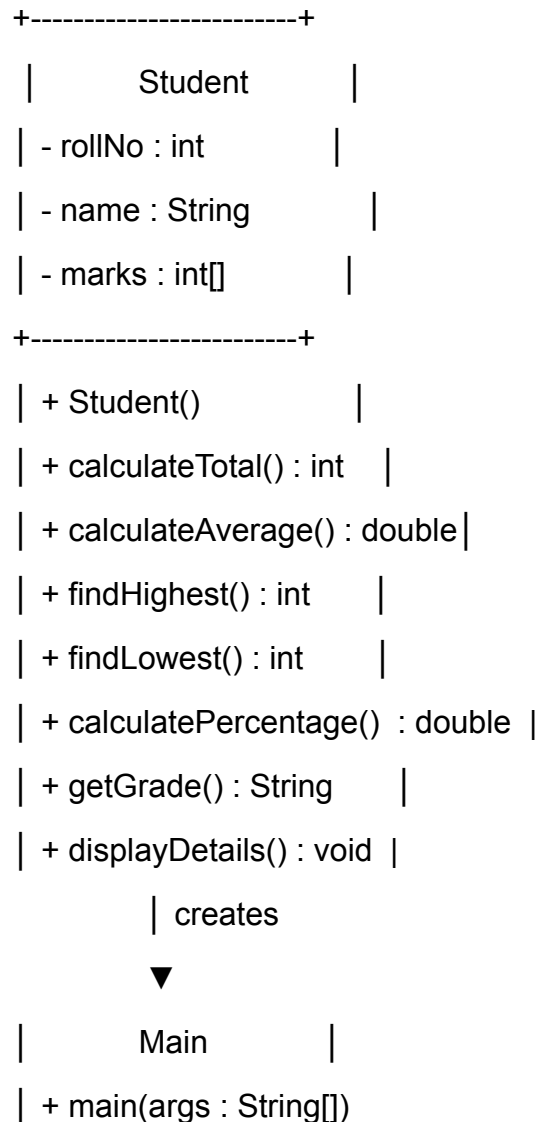
- Computer/Laptop
- Minimum 4 GB RAM
- Keyboard and Monitor

4. Algorithm :

1. Start the program.
2. Create a class named Student.
3. Declare data members for roll number, student name, and marks array.
4. Create a constructor to initialize student details.
5. Store marks of five subjects in an integer array.
6. Calculate the total marks using a loop.
7. Calculate the average marks.
8. Find the highest mark using Math.max().
9. Find the lowest mark using Math.min().
10. Calculate the percentage from the total marks.

11. Round the percentage to two decimal places using Math.round().
12. Process the student's name using trim(), toUpperCase(), and length().
13. Assign a grade based on the percentage.
14. Determine whether the student has passed or failed.
15. Display a performance remark based on the grade.
16. Display the complete student performance report.
17. Stop the program.

5. UML Diagram :



6.Source code :

```
Import java.util.Scanner;
```

```
Class Student {  
    private int rollNo;  
    private String name;  
    private int[] marks;  
    Student(int rollNo, String name, int[] marks) {  
        this.rollNo = rollNo;  
        this.name = name;  
        this.marks = marks;  
    }  
    int calculateTotal() {  
        int total = 0;  
        // Calculate total marks using the array  
        for (int mark : marks) {  
            total += mark;  
        }  
        return total;  
    }  
    double calculateAverage() {  
        return (double) calculateTotal() / marks.length;  
    }  
    int findHighest() {  
        int highest = marks[0];  
        // Find the highest mark  
        for (int mark : marks) {  
            highest = Math.max(highest, mark);  
        }  
        return highest;  
    }  
}
```

```

}

int findLowest() {
    int lowest = marks[0];
    // Find the lowest mark
    for (int mark : marks) {
        lowest = Math.min(lowest, mark);
    }
    return lowest;
}

double calculatePercentage() {
    return ((double) calculateTotal() / (marks.length * 100)) * 100;
}

string getGrade() {
    double percentage = calculatePercentage();
    if (percentage >= 90)
        return "A+";
    else if (percentage >= 80)
        return "A";
    else if (percentage >= 70)
        return "B";
    else if (percentage >= 60)
        return "C";
    else if (percentage >= 50)
        return "D";
    else
        return "F";
}

```

```

String getRemark() {
    String grade = getGrade();
    switch (grade) {
        case "A+":
            return "Outstanding Performance";
        case "A":
            return "Excellent Performance";
        case "B":
            return "Very Good Performance";
        case "C":
            return "Good Performance";
        case "D":
            return "Satisfactory Performance";
        default:
            return "Needs Improvement";
    }
}

void displayDetails() {
    String formattedName = name.trim().toUpperCase();
    // Round percentage to two decimal places
    double percentage = Math.round(calculatePercentage() * 100.0) / 100.0;
    System.out.println("\n----- STUDENT PERFORMANCE REPORT -----");
    System.out.println("Roll Number    : " + rollNo);
    System.out.println("Student Name    : " + formattedName);
    System.out.println("Name Length     : " + formattedName.length());
    System.out.println("Total Marks     : " + calculateTotal());
    System.out.println("Average Marks   : " + calculateAverage());
}

```

```

        System.out.println("Highest Mark    : " + findHighest());
        System.out.println("Lowest Mark     : " + findLowest());
        System.out.println("Percentage     : " + percentage + "%");
        System.out.println("Grade         : " + getGrade());
        if (percentage >= 50)
            System.out.println("Result      : PASS");
        else
            System.out.println("Result      : FAIL");
        System.out.println("Remark       : " + getRemark());
    }
}

public class StudentPerformanceAnalysis {
    public static void main(String[] args) {
        scanner sc = new Scanner(System.in);
        System.out.print("Enter Roll Number: ");
        int rollNo = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Student Name: ");
        String name = sc.nextLine();
        int[] marks = new int[5];
        System.out.println("Enter marks for 5 subjects:");
        for (int i = 0; i < marks.length; i++) {
            System.out.print("Subject " + (i + 1) + ": ");
            marks[i] = sc.nextInt();
        }

        Student student = new Student(rollNo, name, marks);
    }
}

```

```

        Student.displayDetails();

        sc.close();
    }
}

```

6. Execution and Compilation Commands :

Save the program as: StudentPerformanceAnalysis.java

Compile the program: Javac StudentPerformanceAnalysis.java

Run the program: Java StudentPerformanceAnalysis

8. Output screenshot:

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\anits\Desktop\Java-Projects> javac StudentPerformanceAnalysis.java
PS C:\Users\anits\Desktop\Java-Projects> java StudentPerformanceAnalysis
Enter Roll Number: 101
Enter Student Name: John Doe
Enter marks for 5 subjects:
Subject 1: 85
Subject 2: 78
Subject 3: 92
Subject 4: 88
Subject 5: 76

----- STUDENT PERFORMANCE REPORT -----
Roll Number      : 101
Student Name     : JOHN DOE
Name Length      : 8
Total Marks      : 419
Average Marks    : 83.8
Highest Mark     : 92
Lowest Mark      : 76
Percentage       : 83.8%
Grade            : A
Result           : PASS
Remark           : Excellent Performance
PS C:\Users\anits\Desktop\Java-Projects>

```

9. Result :

The Student Performance Analysis System was successfully implemented in Java using Object-Oriented Programming concepts, arrays, String operations, and built-in Math methods. The program successfully calculates and displays the student's total marks, average, highest and lowest marks, percentage, grade, pass/fail result, and performance remark.